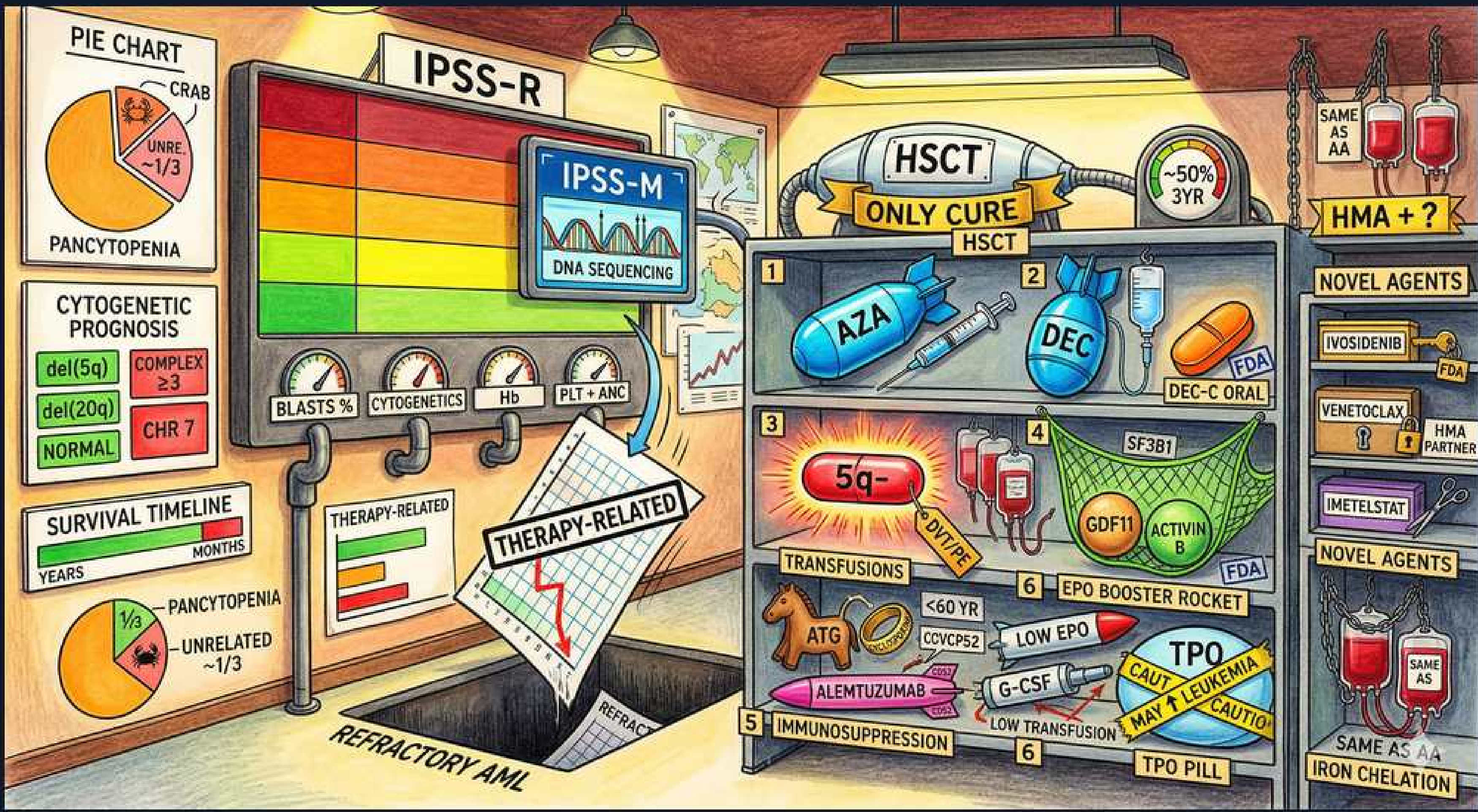
MDS — The MDS Scoreboard & Arsenal (Prognosis & Treatment)



1. IPSS-R scoreboard

WHAT YOU SEE

Red-to-green 'IPSS-R' risk meter on the wall

WHAT IT ENCODES

The revised IPSS-R is the standard MDS prognostic tool, scoring five variables — marrow blast %, cytogenetic risk group, hemoglobin, platelets, and ANC — into five risk tiers (very-low, low, intermediate, high, very-high) with markedly different median survivals (years for very-low vs <1 year for very-high). The practical split is LOWER-risk (very-low/low, sometimes intermediate) → treat the cytopenias; HIGHER-risk → alter natural history with HMAs/transplant.

BOARD TRAP

Two patients with the same blast count get the same drug. WRONG answer: treat by blast count or symptoms alone. Discriminator: therapy is RISK-ADAPTED by IPSS-R — lower-risk MDS is managed with ESAs/luspatercept/lenalidomide/supportive care to improve cytopenias, while higher-risk gets azacitidine and transplant evaluation; conflating the two is the core MDS treatment error.

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2. IPSS-M / DNA sequencing

WHAT YOU SEE

'IPSS-M' monitor labeled DNA sequencing

WHAT IT ENCODES

IPSS-M (molecular, 2022) refines IPSS-R by adding ~31 gene mutations. Adverse-weighted mutations include TP53 (multi-hit), ASXL1, RUNX1, EZH2, NRAS, and FLT3; SF3B1 (without other adverse features) is favorable. It reclassifies a substantial fraction of patients up or down a risk tier versus IPSS-R, sharpening transplant timing decisions.

BOARD TRAP

A patient is 'low-risk' by IPSS-R, so transplant is deferred indefinitely. WRONG answer: rely on cytogenetics-only scoring when molecular data exist. Discriminator: an adverse mutation profile (e.g., multi-hit TP53, ASXL1/RUNX1/EZH2) can upstage an apparently low-risk IPSS-R patient on IPSS-M — molecular testing can change the recommendation toward earlier transplant.

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3. Blasts / cytogenetics / Hb / PLT+ANC

WHAT YOU SEE

Gauges: BLASTS %, CYTOGENETICS, Hb, PLT+ANC

WHAT IT ENCODES

These are the five IPSS-R inputs. Marrow blast % (≤2 / >2–<5 / 5–10 / >10) and cytogenetic group carry the most weight; hemoglobin, platelets, and ANC capture cytopenia depth. Higher blasts and adverse cytogenetics dominate the score and predict AML transformation, while the cytopenia thresholds (e.g., Hb <8, platelets <50, ANC <0.8) modulate it.

BOARD TRAP

Vignette adds CRAB/end-organ criteria into the MDS score. WRONG answer: include calcium/renal/lytic-lesion criteria. Discriminator: IPSS-R uses ONLY blasts, cytogenetics, Hb, platelets, and ANC — CRAB belongs to myeloma; importing it into MDS prognostication is a distractor.

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4. Cytogenetic risk (del5q / complex / chr7)

WHAT YOU SEE

Cards: del(5q), complex ≥3, CHR 7, NORMAL

WHAT IT ENCODES

Cytogenetics anchor prognosis: isolated del(5q) and −Y are very-good/good; normal karyotype is good; monosomy 7/del(7q) is poor; and complex karyotype (≥3 abnormalities, especially with TP53) is very-poor. The five IPSS-R cytogenetic groups independently predict both survival and time to AML.

BOARD TRAP

Any chromosome-5 finding labeled favorable. WRONG answer: assume −5 in a complex karyotype is good prognosis. Discriminator: ISOLATED del(5q) is a GOOD-prognosis, lenalidomide-responsive subtype, but −5 within a complex (≥3) karyotype and monosomy 7 are HIGH-risk — never lump them together.

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5. Pancytopenia / CRAB

WHAT YOU SEE

Pie chart with CRAB and pancytopenia wedges

WHAT IT ENCODES

In MDS, the depth and number of cytopenias drive symptoms (fatigue/anemia, infection from neutropenia, bleeding from thrombocytopenia) and feed the IPSS-R. The CRAB acronym (hyperCalcemia, Renal failure, Anemia, Bone lesions) is the end-organ criteria for symptomatic MULTIPLE MYELOMA, not MDS, and is a common look-alike distractor.

BOARD TRAP

Question lists 'CRAB criteria met' as evidence of high-risk MDS. WRONG answer: use CRAB to risk-stratify MDS. Discriminator: CRAB defines myeloma end-organ damage; MDS prognosis is scored by cytopenias, blast %, and cytogenetics (IPSS-R) — seeing CRAB should redirect you to a plasma-cell disorder, not MDS staging.

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6. Therapy-related MDS

WHAT YOU SEE

'THERAPY-RELATED' falling chart into a trapdoor

WHAT IT ENCODES

Therapy-related MDS (post alkylator/radiation or topoisomerase-II inhibitor) is biologically aggressive — enriched for TP53 mutations and complex/−5/−7 karyotypes — with poor response to chemotherapy and a high rate of AML transformation. It generally falls into higher IPSS-R/M tiers and favors early transplant evaluation when the patient is eligible.

BOARD TRAP

t-MDS managed the same as de novo low-risk MDS with watchful waiting. WRONG answer: treat t-MDS as ordinary low-risk disease. Discriminator: therapy-related MDS carries an inherently worse prognosis driven by TP53/complex karyotype — it should be staged aggressively and referred for transplant rather than observed like de novo lower-risk MDS.

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7. Refractory AML

WHAT YOU SEE

'REFRACTORY AML' on the floor

WHAT IT ENCODES

Higher-risk MDS progresses to secondary AML in a large share of patients. AML arising from antecedent MDS ('secondary/AML-MR' with MDS-related cytogenetics or mutations) responds poorly to conventional 7+3 induction and has inferior survival compared with de novo AML; CPX-351 (liposomal daunorubicin/cytarabine) and HMA+venetoclax are preferred in eligible patients.

BOARD TRAP

Secondary AML from prior MDS treated with standard 7+3 with expectation of de novo cure rates. WRONG answer: expect typical AML outcomes. Discriminator: AML evolving from MDS is chemo-refractory with MDS-related genetics — CPX-351 (fit patients) or azacitidine+venetoclax (unfit) outperforms conventional induction, and transplant is the goal for durable control.

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8. HSCT — only cure

WHAT YOU SEE

'HSCT ONLY CURE' banner, ~50% 3yr

WHAT IT ENCODES

Allogeneic hematopoietic stem-cell transplant is the ONLY potentially curative therapy for MDS, acting through both myeloablation and a graft-versus-leukemia effect, with roughly ~30–50% long-term disease-free survival depending on risk. It is reserved for higher-risk, transplant-eligible patients (fitness, donor, organ function); HMAs may be used as a bridge to transplant.

BOARD TRAP

A higher-risk, fit 60-year-old is offered azacitidine as definitive 'cure.' WRONG answer: call HMAs or supportive care curative. Discriminator: ONLY allogeneic transplant cures MDS — HMAs and ESAs modify disease/symptoms but are not curative, so an eligible higher-risk patient should be referred for transplant rather than left on lifelong HMA.

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9. Azacitidine / decitabine (HMA)

WHAT YOU SEE

AZA syringe and DEC pill (HMA = ?) in the arsenal

WHAT IT ENCODES

Hypomethylating agents (HMAs) — azacitidine and decitabine — are first-line for HIGHER-risk MDS not proceeding immediately to transplant. They are cytidine-analog DNMT inhibitors that hypomethylate and re-express silenced genes. Azacitidine (75 mg/m²/day SC ×7 days q28d) improved overall survival vs conventional care in higher-risk MDS; responses take 4–6 cycles, so don't stop early. Oral decitabine/cedazuridine is also available.

BOARD TRAP

Patient deemed an HMA non-responder after 2 cycles and switched off. WRONG answer: abandon the HMA prematurely. Also WRONG: presenting HMAs as curative. Discriminator: HMAs require ≥4–6 cycles for response and are disease-modifying/survival-prolonging but NOT curative — they bridge or palliate, never replace allogeneic transplant in eligible higher-risk patients.

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10. Lenalidomide for del(5q)

WHAT YOU SEE

'5q-' card on the shelf

WHAT IT ENCODES

Lenalidomide is the treatment of choice for transfusion-dependent anemia in LOWER-risk MDS with isolated del(5q) (5q− syndrome). At 10 mg/day (21 of 28 days) it achieves transfusion independence in ~60–70% and cytogenetic responses in many by selectively suppressing the del(5q) clone (via CK1α/casein-kinase degradation). Monitor for neutropenia/thrombocytopenia early in therapy.

BOARD TRAP

Low-risk isolated-del(5q) anemia started on azacitidine first-line. WRONG answer: reflexively use an HMA for low-risk del(5q). Discriminator: del(5q) → lenalidomide (after/instead of ESA depending on EPO level and transfusion burden); HMAs are for higher-risk disease, not low-risk isolated del(5q).

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11. Luspatercept / SF3B1

WHAT YOU SEE

Net labeled SF3B1 / GDF11 / activin (EPO booster)

WHAT IT ENCODES

Luspatercept is an erythroid-maturation agent (activin-receptor-IIB ligand trap) that sequesters TGF-β-superfamily ligands (GDF11/activin) to promote late-stage erythroid maturation. It treats anemia in lower-risk MDS, with the strongest benefit in ring-sideroblast/SF3B1-mutated disease; it is now used first-line for lower-risk anemia in many patients (COMMANDS data) regardless of ring sideroblasts.

BOARD TRAP

Anemic lower-risk patient with high serum EPO and transfusion dependence given another ESA. WRONG answer: keep pushing ESAs when EPO is already high. Discriminator: luspatercept works distinct from EPO (it boosts terminal erythroid maturation, not progenitor proliferation) and is the preferred choice for ring-sideroblast/SF3B1 anemia and for ESA-unsuitable patients — not a redundant ESA.

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12. EPO (ESA) for low-risk anemia

WHAT YOU SEE

EPO booster rocket, 'LOW EPO <500'

WHAT IT ENCODES

Erythropoiesis-stimulating agents (epoetin/darbepoetin) are first-line for anemia in LOWER-risk MDS, working best when endogenous serum EPO is LOW (<500 mU/mL, ideally <200) and the transfusion burden is light (<2 units/month). Adding G-CSF can synergize for erythroid response, particularly in ring-sideroblast subtypes.

BOARD TRAP

Transfusion-dependent lower-risk patient with serum EPO 900 started on epoetin. WRONG answer: use an ESA when endogenous EPO is high. Discriminator: ESA efficacy depends on a LOW baseline EPO (<500) and low transfusion need — with high EPO or heavy transfusion dependence, response is poor and luspatercept (or lenalidomide for del(5q)) is preferred.

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13. Immunosuppression (ATG)

WHAT YOU SEE

'IMMUNOSUPPRESSION' ATG, <60 yr, hypocellular

WHAT IT ENCODES

Immunosuppressive therapy — antithymocyte globulin (ATG) ± cyclosporine — benefits a selected minority of LOWER-risk MDS that behaves like aplastic anemia: younger patients (<60), HYPOcellular marrow, low blasts, normal cytogenetics, PNH clone, and/or HLA-DR15 positivity. It targets the T-cell-mediated marrow suppression in hypoplastic MDS.

BOARD TRAP

Older patient with hypercellular high-risk MDS offered ATG. WRONG answer: use immunosuppression broadly in MDS. Discriminator: ATG/cyclosporine is reserved for the AA-like, HYPOcellular, low-risk, younger subset — it has no role in typical hypercellular or higher-risk MDS, which need HMAs/transplant.

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14. Transfusions + iron chelation

WHAT YOU SEE

Blood bags / 'IRON CHELATION' on the right

WHAT IT ENCODES

Supportive care is the backbone for many patients: RBC transfusions for symptomatic anemia (with leukoreduced/irradiated products to limit alloimmunization and TA-GVHD, especially pre-transplant) and platelets for bleeding. Chronic RBC transfusion (often >20–25 units or rising ferritin >1000–2500) causes iron overload; chelation (deferasirox) is used in transfusion-dependent LOWER-risk patients with reasonable life expectancy and transplant candidates.

BOARD TRAP

Prophylactic platelet transfusions and broad iron chelation for every MDS patient. WRONG answer: chelate everyone and transfuse platelets prophylactically without thresholds. Discriminator: platelets are given for bleeding or below threshold (not routinely for all), and chelation is selective — for transfusion-dependent LOWER-risk/transplant-eligible patients, not short-prognosis higher-risk disease where it offers no benefit.

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15. Novel agents (ivosidenib, venetoclax, imetelstat)

WHAT YOU SEE

'NOVEL AGENTS' shelf — ivosidenib, venetoclax, imetelstat

WHAT IT ENCODES

Targeted/emerging options: ivosidenib (IDH1 inhibitor) and enasidenib (IDH2) for IDH-mutated MDS; venetoclax (BCL-2 inhibitor) combined with HMAs for higher-risk MDS/secondary AML; and imetelstat (telomerase inhibitor) for transfusion-dependent LOWER-risk MDS refractory to ESAs. These extend the arsenal beyond HMAs and supportive care but are not curative substitutes for transplant.

BOARD TRAP

IDH inhibitor given to an MDS patient without testing IDH mutation status. WRONG answer: use targeted agents empirically. Discriminator: ivosidenib/enasidenib require a documented IDH1/IDH2 mutation, and imetelstat is for ESA-refractory transfusion-dependent LOWER-risk disease — these are biomarker- and risk-specific, not one-size-fits-all, and still do not replace allogeneic transplant for cure.
